

Syllabus

Department	Math & CS	Credits	3	
Class Room	Changui-gwan 5701	Subject	MC0000:	Reinforcement
			Learning	
Instructor	Sumin Han	Class	1	
Lab (e-mail)	<code>suminhan@ksa.hs.kr</code>	hrs/wk	3	

1. Course Description

Course Objectives

On completion of this course, you'll hopefully be able to:

- Formalize a decision-making problem as a Markov Decision Process (MDP) and derive the Bellman equations.
- Implement and compare tabular reinforcement learning algorithms: dynamic programming, Monte Carlo methods, and temporal-difference learning.
- Understand and implement deep reinforcement learning methods that combine function approximation with neural networks (DQN, policy gradient methods, PPO).
- Understand the principles behind modern applications such as imitation learning and RLHF (Reinforcement Learning from Human Feedback).
- Apply reinforcement learning algorithms to a simulated environment as a team, following an honest experimental protocol (multiple seeds, baseline comparison).

Contents

- Introduction to reinforcement learning and a neural network review
- Multi-armed bandits
- Markov Decision Process (MDP) formalization
- Dynamic programming
- Monte Carlo methods
- Temporal-difference learning
- n -step bootstrapping, eligibility traces, and planning
- **[Block A Capstone] Team project: tabular reinforcement learning**
- Function approximation and DQN
- Policy-based reinforcement learning
- Advanced policy optimization: PPO
- Imitation learning and RLHF
- Robot simulation and control basics
- Advanced simulation: MuJoCo and Isaac Sim
- Model-based RL and Monte Carlo Tree Search (MCTS)
- **[Block B Capstone] Team project: deep reinforcement learning**

This course is a follow-up to “Machine Learning & Deep Learning” and assumes prior familiarity with neural networks.

2. Text & References

Course materials: instructor-prepared lecture notes and slides.

- “Reinforcement Learning: An Introduction” (2nd ed.), Richard S. Sutton and Andrew G. Barto

3. Grading

Activities	Quizzes	Homework	Midterm	Final	Team Project	Attendance
Percentages	10%	10%	25%	25%	25%	5%

- **Quizzes:** a short True/False quiz each week checking conceptual understanding of that week’s material.
- **Homework:** individual programming assignments implementing the reinforcement learning algorithms covered in class.
- **Midterm / Final:** written exams covering the theory of Block A and Block B respectively.
- **Team Project:** two capstone projects (Week 8: tabular RL, Week 16: deep RL), each graded on a written report, a presentation, and peer-review. The grading emphasizes the *honesty of the experimental protocol* (seeds ≥ 3 , baseline comparison against a random policy, honest reporting of unstable results) over raw performance.
- **Make-up exam:** available for officially approved absences (e.g. illness, out-of-school competition).
- **Late-work policy:** not accepted unless officially approved.

4. Lecture Schedule

Week	Topic	Note
1	Introduction to RL, Neural Network Review	
2	Multi-Armed Bandits	
3	MDP Formalization	
4	Dynamic Programming	
5	Monte Carlo Methods	
6	Temporal-Difference Learning	
7	n -step Bootstrapping, Eligibility Traces, Planning	
8	Block A Capstone: Team Project	Midterm exam
9	Function Approximation and DQN	
10	Policy-Based Reinforcement Learning	
11	Advanced Policy Optimization: PPO	
12	Imitation Learning and RLHF	
13	Robot Simulation and Control Basics	
14	Advanced Simulation: MuJoCo, Isaac Sim	
15	Model-Based RL and Monte Carlo Tree Search	
16	Block B Capstone: Team Project	Final exam

The exact calendar (start date, midterm/final exam period, holidays) will be announced once the official 2027 academic calendar is confirmed.